

WASTE NO MORE TIME ARGUING ABOUT
WHAT A GOOD MAN SHOULD BE. BE ONE.

- MARCUS AURELIUS



Agenda

- ① Salting, Repartition and coalesce
- ② Adaptive Query Execution [AQE]
- ③ Advanced configs
- ④ Interview question

Repartition

Making sure to split up or shuffle data again
based on input parameters

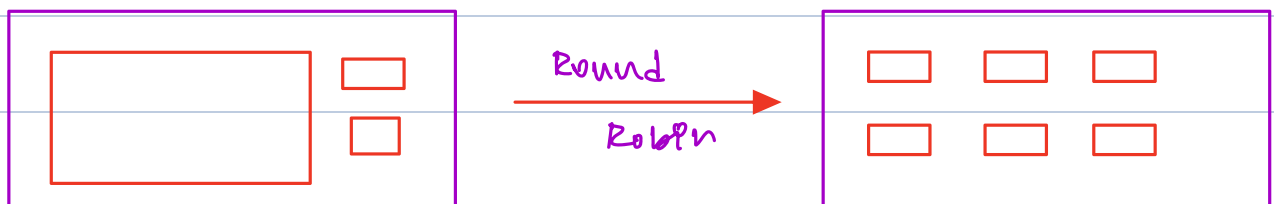
Repartitions $\rightarrow$ num of partitions

$\rightarrow$ column to partition

used for Precreating number of partitions

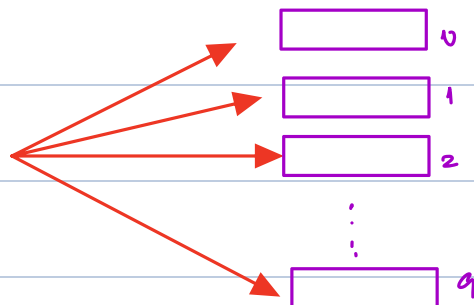
wide transformation

large DF. repartition (200)



large DF. repartition (10, "id")

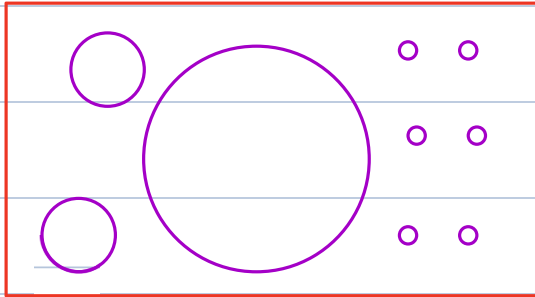
$\text{Hash}(\text{id}) \% 10$



Hash collection

Coalesce

Merging 2 or more partitions into single partition

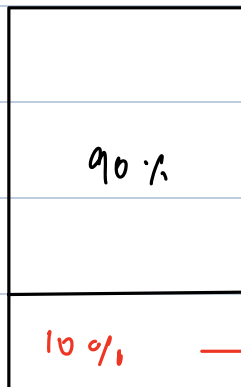


abhiik , saurav , milan
armaan , Dhruv

so we are performing lesser amount of shuffle

coalesce is faster compared to repartition
Narrow transformation

Salting



one key

Company USA 90%

Europe, Asia, Africa
10%

other key

In this case as

you perform any wide transformation on
country, 90% data will be present in

1 partition → same value will move
to same partition

Salting as a concept.

USA	USA_0
USA	USA_1
USA	USA_2
USA	USA_3
India	India_0

adding random values at
end of key to split
data almost equally

① my original key is
not being worked on

② How much distinct values
I need to add

SALT FACTOR determines how much uniqueness
you need to add

Figure out the value using trial & error
method taking guidance from **partition size**
spark v1

Task → get count of country → DF

USA	USA_0
USA	USA_1
USA	USA_0
USA	USA_1
India	India_0

groupBy
salted

USA_0	2
USA_1	2
India_0	1

remove
salt & perform
groupBy.

Benefit

- ① why salt?
- ② How it helps?

USA	4
India	1

Compare

① old

② New

5 / DS

23 / DS

count of country.

90% in one go unoptimized

40% in one go → Intermediate ans

↓ Join

Adaptive Query Execution



next class